

Mortal Computation

Mortality, the Genome, and Life-History in Cost-Accounted GSLTs

L. G. Meredith

F1R3FLY.io

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Abstract

We read the cost-accounting (metering) construction on graph-structured lambda theories (GSLTs) as a biology rather than as an economics. The token-gated rewrite that meters computation is, on the nose, a mechanism for making computation *mortal*: death is deadlock-by-starvation, and metabolism is the splitter/joiner token economy that postpones it. For Turing-complete continued-interactive GSLTs the metered calculus *internalises* into the original model up to weak bisimulation—cost accounting is a definitional extension over a universal base—and the image of that internalisation is a sublanguage in which mortality and metabolism are implemented from native parts. Turing completeness further yields, by composing the two universality maps, a meta-circular interpreter; when the model hosts reflection natively this interpreter is internalised, giving the quote/dequote pair the dual-use character von Neumann required of a self-reproducing description. Quoted description behaves as a potentially immortal germ line; the dequoted running process is the mortal soma. With reproduction thereby isolated as a metabolic cost separable from a computation's other functions, we recover life-history theory: the logistic law $\dot{N} = rN(1 - N/K)$ emerges as the mean-field of a population of cost-accounted replicators on a bounded-inflow substrate, with r and K as closed forms in the construction's primitives, and r - versus K -selection appears as the two extreme vertices of a continuous allocation simplex.

1 Setup: the cost monad and its internalisation

Write an interactive GSLT as $\mathcal{G} = (\Sigma, E, R)$ with a distinguished interacting sort carrying a labelled transition system, structural equations E , and rewrite rules R . The cost construction [2] is an endofunctor $\mathbf{C}$ on **ciGSLT**, the category of *continued* interactive GSLTs and bisimulation-preserving morphisms—continued because the temporal token stack must drain in step with reduction, which requires a calculus whose interaction is presented as a meterable cut, a proper structural enrichment of bare interactivity. On objects $\mathbf{C}$ adjoins a signature sort $\mathbf{Sig}$ as a commutative monoid, a token-stack sort $S ::= () \mid s : S$ as the free $\mathbf{Sig}$ -graded structure, a signed-term wrapper $\{\cdot\}_s$, and a constructor **sig-of** that mints a fresh signature from runtime data. It replaces each rule $\ell \rightarrow r$ of R by its *gated family*: the redex fires only when matching tokens are present and consuming them. The five rules that govern signed communication are not five facts; they are the image of the single communication rule under $\mathbf{C}$ [1].

$\mathbf{C}$ is moreover a *monad* $(\mathbf{C}, \eta, \mu)$ on **ciGSLT** [2]. The unit $\eta_{\mathcal{G}}(P) = \{P\}_()$ wraps a term with the unit signature against a freely available unit token, embedding $\mathcal{G}$ as the *cost-free fragment* of $\mathbf{C}\mathcal{G}$ —the metered calculus restricted to trivially-funded interactions, where every redex fires

without net resource. The multiplication $\mu_{\mathcal{G}} : \mathcal{C}^2\mathcal{G} \rightarrow \mathcal{C}\mathcal{G}$ flattens metered metering by multiplying nested signatures and concatenating stacks-of-stacks. This monad is *not* idempotent: μ is no isomorphism, because concatenation forgets the boundary between two stacks and signature-multiplication forgets the factorisation; flattening is a genuine merge of two resource accounts, not the collapse of a redundant copy [2]. The non-idempotence will matter in §5.

We assume throughout that $\mathcal{G}$ is Turing-complete and *continued-interactive*: it can express persistent processes that continue to interact after producing output. Under this assumption the entire metering apparatus of $\mathcal{C}\mathcal{G}$ —signature equality, stack maintenance, gating—is computable, hence encodable in $\mathcal{G}$ itself, and $\mathcal{C}\mathcal{G}$ admits an *internalisation*

$$\text{Imp}_{\mathcal{G}} : \mathcal{C}\mathcal{G} \longrightarrow \mathcal{G},$$

a weak simulation realising the cost-accounted semantics inside the un-metered but universal base [2]: signatures map to channel names (via reflection where available, or to fresh names otherwise), tokens become messages on those channels carrying continuation payloads, and signed processes become *fuel-gates*—guard patterns that must consume a token message before the underlying redex proceeds. Splitters and joiners reconcile combined and decomposed token granularities. The internalisation is *behaviour-preserving but structure-dissolving*: the clean sorts Sig, S dissolve into encoded data, and the agreement holds up to *weak* bisimulation, since the simulating interpreter interposes administrative reductions.

Crucially, $\text{Imp}_{\mathcal{G}}$ is not an isolated map but the right half of an *adjoint retraction* $\eta_{\mathcal{G}} \dashv \text{Imp}_{\mathcal{G}}$ in the bicategory of continued interactive GSLTs and simulations: $\text{Imp}_{\mathcal{G}} \circ \eta_{\mathcal{G}} \approx_w \text{id}_{\mathcal{G}}$, so $\mathcal{C}\mathcal{G}$ is a behavioural retract of $\mathcal{G}$ up to weak bisimulation, and cost accounting is a definitional extension over the universal base [2]. The object of study here is the image $\text{im}(\text{Imp}_{\mathcal{G}}) \subseteq \mathcal{G}$, a distinguished sublanguage built entirely from native constructs. The claim of this note is that the biology—mortality, the genome, and life-history—lives in that image, and that all of it is therefore native to any Turing-complete continued-interactive $\mathcal{G}$, the cost construction merely locating where. Because $\eta_{\mathcal{G}} \dashv \text{Imp}_{\mathcal{G}}$ is an adjoint retraction, that image is a well-defined behavioural retract rather than a construction artifact—a point we return to in §5.

2 Mortality and metabolism

Death. In $\mathcal{C}\mathcal{G}$ a redex fires only by consuming a matching token; a process whose stack is $()$ has no enabled rewrite and blocks permanently. Pushing this through $\text{Imp}_{\mathcal{G}}$, the token message on the relevant signature-channel simply stops arriving and the fuel-gate never completes. In $\mathcal{G}$ this is an ordinary configuration: a guarded process whose guard is never discharged.

Definition 1 (Starvation death). A process P in $\text{im}(\text{Imp}_{\mathcal{G}})$ is *dead* at a configuration if every redex of P is fuel-gated and no token message is reachable on any of its gating channels. Death is permanent deadlock by fuel exhaustion.

Mortality is thus not a new primitive but a fragment of $\mathcal{G}$ that $\mathcal{G}$ admitted all along; $\mathcal{C}$ exhibits which fragment.

Metabolism. The token economy that postpones death is metabolic in a precise sense. A joiner assembles a high-denomination token from components (anabolism); a splitter cleaves one into spendable units (catabolism); and the gated rewrite couples each unit of computational work to a unit of fuel consumption—no step without an accompanying expenditure. This is

the same coupling that defines a metabolism thermodynamically. Read against a Ruelle–Bowen potential [11, 3] on the transition graph, the splitter/joiner dynamics are the flow and the zero-fuel configurations are the absorbing set: metabolism is motion on the landscape, death is its absorbing boundary.

The cost-free fragment, and the Buss argument. A parallel composition of metered processes is exactly the kind of population that supports selection: copies are autonomous ($P \mid P \not\equiv P$ once embedded in distinct local contexts), and selection exploits inexactness rather than collapsing it. The un-metered $\mathcal{G}$ sits inside $\mathcal{CG}$ as the cost-free fragment $\eta_{\mathcal{G}}(\mathcal{G})$, where every interaction is funded at unit cost and nothing can starve; this is the home of the immortal replicator. There a $!P$ with a faster-cycling P always invades a parallel composition without bound; no configuration is stable against its own fastest subterm. This is Buss’s cell-lineage problem [4] in its rawest form. Leaving the cost-free fragment for a genuinely finitely-funded process—an element of $\mathcal{CG} \setminus \eta_{\mathcal{G}}(\mathcal{G})$, internalised by $\text{Imp}_{\mathcal{G}}$ into $\text{im}(\text{Imp}_{\mathcal{G}})$ —bounds the defector: it out-cycles its neighbours only until its stack is spent, after which it deadlocks. Mortality is precisely the act of leaving the cost-free fragment.

Proposition 1 (Mortality caps invasion). *Let D be a subterm-variant with per-cycle fuel draw strictly exceeding its per-cycle fuel income on a shared bounded substrate. In $\mathcal{CG}$, D executes for a number of cycles bounded by its initial endowment and then dies, so the cumulative interactions it contributes are finite. In the un-metered $\mathcal{G}$ the same D contributes unboundedly. Hence configurations in $\text{im}(\text{Imp}_{\mathcal{G}})$ that sequester the minting constructor **sig-of** admit no heritable take-over by a high-draw somatic defector, whereas no configuration of the cost-free fragment $\eta_{\mathcal{G}}(\mathcal{G})$ resists one.*

The biological reading is Buss’s: somatic mortality together with germ sequestration aligns the levels of selection by denying a within-individual defector any future. Mortality is selected not in spite of bounding individual runs but because a bounded run is the precondition for a stable higher-level individual that can itself be a durable unit of selection.

3 Reflection and the genome

A self-interpreter from universality. Turing completeness furnishes two universality maps. The first, $e : \text{TM} \rightarrow \mathcal{G}$, is completeness proper: $\mathcal{G}$ simulates any machine. The second, $i : \mathcal{G} \rightarrow \text{TM}$, is the Church–Turing direction [13]: any effectively presented reduction relation can be run by a machine. Their composite

$$i \circ e : \mathcal{G} \longrightarrow \mathcal{G}$$

is a program *in* $\mathcal{G}$ that takes a $\mathcal{G}$ -program as data and runs it, correct up to weak bisimulation: a meta-circular interpreter. It is the same internalisation that §1 used to realise $\mathcal{CG}$ inside $\mathcal{G}$ —an instance of $\text{Imp}_{\mathcal{G}}$ specialised to the universal apparatus—and so inherits the administrative reductions that make the agreement weak rather than strong. To feed a program to this interpreter one must first represent it as a value, and the TM detour manufactures such a code-as-data representation with no reflective primitive in sight. *Reflection appears as a consequence of universality*—a theorem, not a term. This is exogenous reflection.

What native reflection changes. When $\mathcal{G}$ hosts reflection natively—as the rho calculus does [8], a reflective higher-order extension of Milner’s π -calculus [9], where $@P$ is already a value in the same algebra as the running process—the meta-circular interpreter is internalised, and the gain is representational, not merely a saved detour. A TM-tape encoding has a brutal mutation neighbourhood: almost any edit to the tape is catastrophic because the tape does not respect the term structure it encodes. A quoted term $@P$ has a benign one: mutation is a structural edit of a subterm and recombination is the splicing of quoted subprocesses, both staying inside the model’s own grammar. Native reflection is therefore the difference between a genome that is evolvable and one that is not.

Proposition 2 (Two readings agree on behaviour, differ on the description). *The exogenous interpreter $i \circ e$ and the internalised interpreter agree up to weak bisimulation on the behaviour of every interpreted program, but differ on the algebra of the code-as-data representation: the former carries a foreign encoding closed under tape edits, the latter the model’s own term algebra closed under structural edits and splices. Evolvability is a property of the latter algebra, not of the interpreted behaviour, and so distinguishes the two readings even though they are behaviourally equivalent.*

Von Neumann’s dual use, recovered. A self-reproducing description φ [14] must support two distinct operations: it is *copied uninterpreted* (replication) and *interpreted to build the machinery* (expression). Native reflection supplies both on a single object. A name $@P$ can be transmitted on a channel and received *without* dequoting—replication, the description copied as inert data—or dequoted by $*$ and run—expression, the description read as instructions. Crick’s transcription/translation split [5] and Weismann’s two modes of the germ-line description [16] are these same two operations on $@P$. The meta-circular structure is what closes the loop: the reader is encoded in what it reads.

Germ and soma from quote and dequote. Refining §2: the germ line is the sublanguage that retains **sig-of** *and* carries the quoted description $@P$ —together exactly what is needed to fuel and to instantiate an offspring. The soma is the dequoted running process $*@P$. Weismann’s mortality asymmetry then falls out of the quote/dequote distinction rather than being postulated:

$$\underbrace{*@P}_{\text{soma}} \text{ metabolises and starves; } \quad \underbrace{@P}_{\text{germ}} \text{ is inert, cannot starve, is re-instantiable.}$$

Mortal soma = the dequoted process. Immortal germ = the quoted description.

Finally, applying a fuel-gate not to a bare communication but to the *dequoting* operation is regulated expression: the gene $@P$ is read and run only when the requisite tokens are present. This is operon logic—expression coupled to metabolic state. The genome’s full operation set is thus native: replication is copying the name, expression is token-gated dequoting, mutation and recombination are edits and splices of the quoted term, and regulation is the fuel-gate deciding *when* the description is read.

4 Life-history theory

Metering makes “the cost of reproduction” a measurable quantity rather than a category error. In the cost-free fragment $\eta_{\mathcal{G}}(\mathcal{G})$, $!P$ replicates for free and nothing is separable from anything else.

Once every operation is token-gated, the reproduction operations—minting via `sig-of`, copying `@P`, endowing the offspring’s initial stack, and the gated dequoting that instantiates it—each carry an identifiable charge in the same currency as every other charge the process incurs; in the rholang realisation these charges are exactly the phlogiston debits of the signed-process fuel-gates [1].

4.1 The metabolic ledger

Per metabolic cycle a replicating computation has realised income B and partitions it into maintenance and reproduction,

$$B = c_{\text{maint}} + c_{\text{rep}}, \quad c_{\text{rep}} \approx n(c_{\text{mint}} + c_{\text{copy}} + e),$$

where n is the number of offspring per cycle and e is the *endowment*—the initial token stack that travels with each copied description `@P`. The endowment is literally metabolic: tokens removed from the parent’s budget and placed in the offspring’s. The partition (maintenance versus reproduction, and within reproduction n versus e) is the life-history allocation problem stated in the construction’s own primitives.

4.2 Carrying capacity as a fuel-balance fixed point and the emergence of the logistic law

Let a bounded token inflow Φ (tokens per unit time) feed the substrate through the splitter/joiner network, shared among N identical replicators. Realised per-process income is density-dependent through contention on the bounded inflow. To leading order we take the contention response to be linear,

$$B(N) = B_0 \left(1 - \frac{N}{N_{\text{sat}}}\right), \quad N_{\text{sat}} = \frac{\Phi}{B_0}, \quad (1)$$

the first-order term of any saturating response that vanishes when aggregate demand NB_0 exhausts the inflow. Per-capita birth rate is the reproductive surplus divided by the unit cost of reproduction, and death has a baseline density-independent rate d_0 (accident and exogenous shock) on top of starvation:

$$b(N) = \frac{\max(0, B(N) - c_{\text{maint}})}{c_{\text{rep}}}, \quad d(N) = d_0. \quad (2)$$

Proposition 3 (Emergence of the logistic law). *Under (1)–(2), the mean-field population dynamics of a set of cost-accounted replicating computations on a bounded-inflow substrate, in the regime $B(N) \geq c_{\text{maint}}$, obey the logistic law [15]*

$$\frac{dN}{dt} = (b(N) - d_0) N = rN \left(1 - \frac{N}{K}\right),$$

with intrinsic rate and carrying capacity given in closed form by the ledger primitives,

$$r = \frac{B_0 - c_{\text{maint}}}{c_{\text{rep}}} - d_0, \quad K = N_{\text{sat}} \left(1 - \frac{c_{\text{maint}} + d_0 c_{\text{rep}}}{B_0}\right) = \frac{\Phi}{B_0} \left(1 - \frac{c_{\text{maint}} + d_0 c_{\text{rep}}}{B_0}\right).$$

Derivation. Substituting (1) into (2) and dropping the clamp in the stated regime,

$$b(N) - d_0 = \frac{B_0(1 - N/N_{\text{sat}}) - c_{\text{maint}}}{c_{\text{rep}}} - d_0 = \underbrace{\left(\frac{B_0 - c_{\text{maint}}}{c_{\text{rep}}} - d_0 \right)}_r - \frac{B_0}{c_{\text{rep}} N_{\text{sat}}} N.$$

Multiplying by N gives $\dot{N} = rN - (B_0/(c_{\text{rep}}N_{\text{sat}}))N^2$, which is $rN(1 - N/K)$ with $K = r c_{\text{rep}}N_{\text{sat}}/B_0$. Expanding r and using $N_{\text{sat}} = \Phi/B_0$ yields the stated closed form. $\square$

Remark. The quadratic term—hence the entire logistic shape—comes from the density-dependence of realised income, i.e. from contention on the bounded inflow. The law is therefore *derived*, not assumed. The qualitative content—carrying capacity set by inflow over per-capita draw, intrinsic rate set by surplus over reproductive unit cost—is robust to the precise contention ansatz; the exact coefficient in K depends on it. In the maintenance-dominated small- d_0 limit the carrying capacity reduces to the bare fuel balance $N^*c_{\text{maint}} = \Phi$, i.e. $K \rightarrow \Phi/c_{\text{maint}}$: as many processes as the inflow can keep from starving. Density-dependent mortality is not imposed; it emerges as the rising starvation rate when Nc_{maint} approaches Φ and the splitters can no longer top every stack.

4.3 r - and K -selection as corners of an allocation simplex

Selection acts on the allocation, and which regime is favoured is set by which term in the ledger binds; the r/K dichotomy [7, 10] is the coarsest reading of it. The criterion is the invasion analysis of §2; only the binding constraint differs.

- **r -corner.** Uncrowded or volatile substrate ($N \ll K$, death dominated by exogenous supply shocks, density-independent) rewards maximising $r = (B_0 - c_{\text{maint}})/c_{\text{rep}} - d_0$: push n up, push e and c_{maint} down. Many minimal-endowment offspring, fire-and-forget, short-lived soma. This is the immortal-replicator defector of §2, now adaptive rather than pathological, because in an empty substrate raw instantiation rate is what invasion rewards.
- **K -corner.** Saturated, stable substrate ($N \approx K$, death dominated by competition, density-dependent) rewards competitive fuel capture: push e and c_{maint} up, push n down. Richly-endowed offspring, long-lived competitive soma; per-offspring capture beats raw count.

Corollaries in the primitives. Parental care is a persistent process that keeps feeding tokens to the offspring’s signature-channel after instantiation—a continued-interactive contract, which is what “continued” is for; r -strategists cut the channel at birth, K -strategists hold it open. Semelparity is the parent dumping its residual budget into one reproductive burst and then deadlocking by design ($c_{\text{maint}} \rightarrow 0$ post-burst); iteroparity retains maintenance to survive further cycles. Body size maps to the soma’s standing token stack—the capital a process keeps on hand—and the familiar r/K body-size correlation appears because large standing capital repays its maintenance draw only when competition, not throughput, binds.

Remark (The construction gives more than the dichotomy). r and K are the two extreme vertices of a continuous allocation simplex over $(c_{\text{maint}}, n, e, \text{timing})$. That continuum is exactly modern age-structured / disposable-soma life-history theory [12, 6], which r/K coarsens into a binary. The construction natively produces the continuum, so the right computational taxonomy of reproduction schemes is a classification of points in the simplex with r and K as named corners—truer to the biology and more natural in the construction than a dichotomy.

5 What to make rigorous next

Three load-bearing claims are stated above at the level a working note allows and deserve full treatment.

1. **Characterising the mortal fragment.** The well-definedness of “the organisms inside $\mathcal{G}$ ” is already secured: $\eta_{\mathcal{G}} \dashv \text{Imp}_{\mathcal{G}}$ is an adjoint retraction, so $\text{im}(\text{Imp}_{\mathcal{G}})$ is a genuine behavioural retract of $\mathcal{G}$, not a construction artifact [2]. Note that the earlier hope of an *idempotent* round trip is foreclosed: $\mathbf{C}$ is a *non-idempotent* monad— μ merges two resource accounts rather than collapsing a redundant copy—so there is no comonadic fixed-point set to appeal to. The genuine open problem is therefore not idempotence but *which* retract-image processes can starve: an internal characterisation of the mortal fragment $\text{Imp}_{\mathcal{G}}(\mathbf{C}\mathcal{G} \setminus \eta_{\mathcal{G}}(\mathcal{G}))$ against the cost-free, hence immortal, fragment $\eta_{\mathcal{G}}(\mathcal{G})$. “The living” is the complement of the cost-free fragment under internalisation; making that complement intrinsic—definable without reference to the apparatus that $\text{Imp}_{\mathcal{G}}$ dissolves—is the central structural question the biological reading rests on.
2. **The fidelity comparison (Prop. 2).** Make precise the sense in which the exogenous and internalised self-interpreters agree on behaviour up to weak bisimulation while differing on the description algebra, and prove that evolvability—benign mutation neighbourhood under structural edits—is a property of the description algebra alone.
3. **The logistic-emergence proposition (Prop. 3).** Promote the mean-field derivation to a theorem: specify the stochastic token-flow process whose hydrodynamic limit is the stated ODE, identify the conditions on the splitter/joiner network under which the contention response (1) is the correct leading order, and characterise the corrections that distinguish the exact K from the bare fuel balance Φ/c_{maint} . This is the technical spine of the programme.

References

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